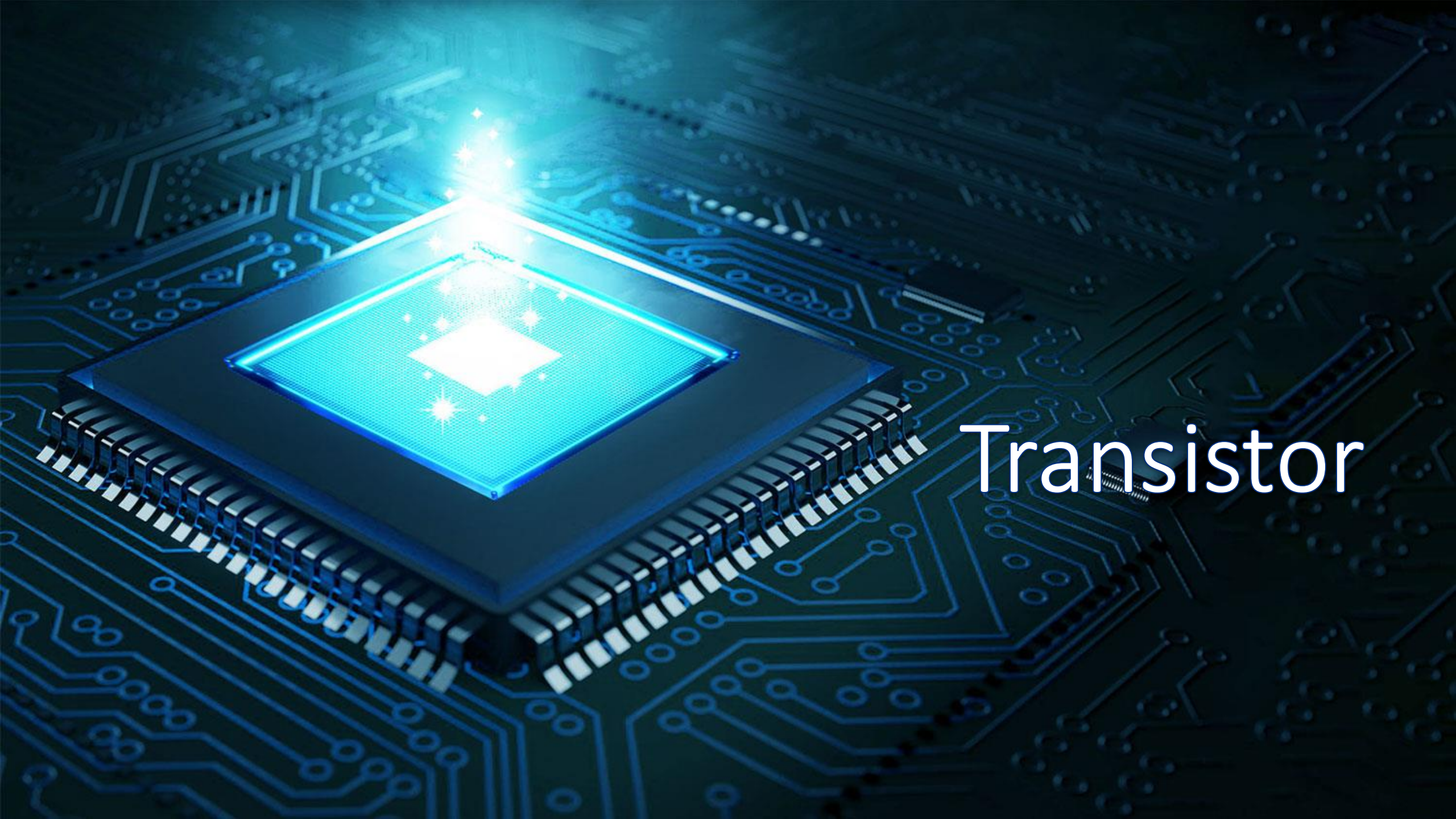
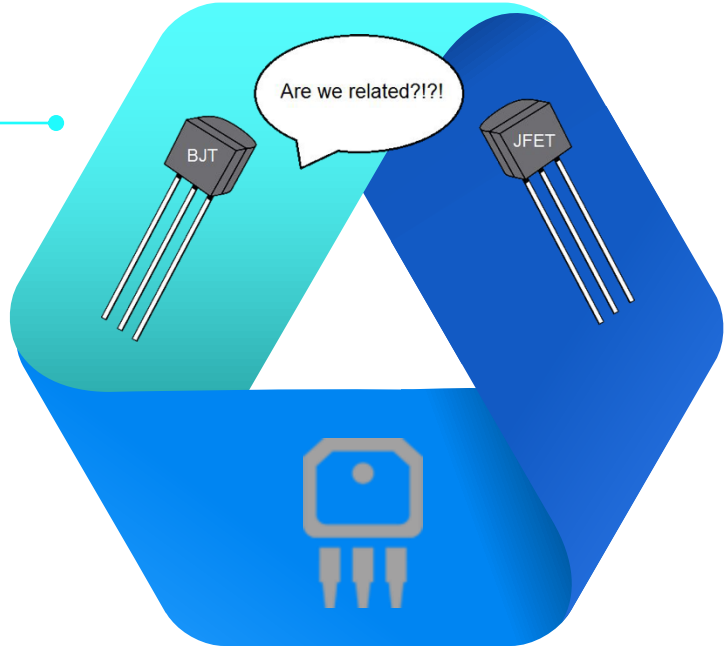

Transistor

What is a transistor?

ADVANTAGES

- compact size
- cheap
- low mechanical sensitivity
- low power consumption
- life span
- high efficiency
- a faster switching rate



DISADVANTAGES

- limited electron mobility
- can't handle any type of interference
- sensitive to temperature
- produced very low energy

Transistors are classified into two major types:
Bipolar junction transistor (BJT)
Junction Field Effect Transistor (JFET)
Transistors have many uses, including amplification, switching, voltage regulation, and the modulation of signals.

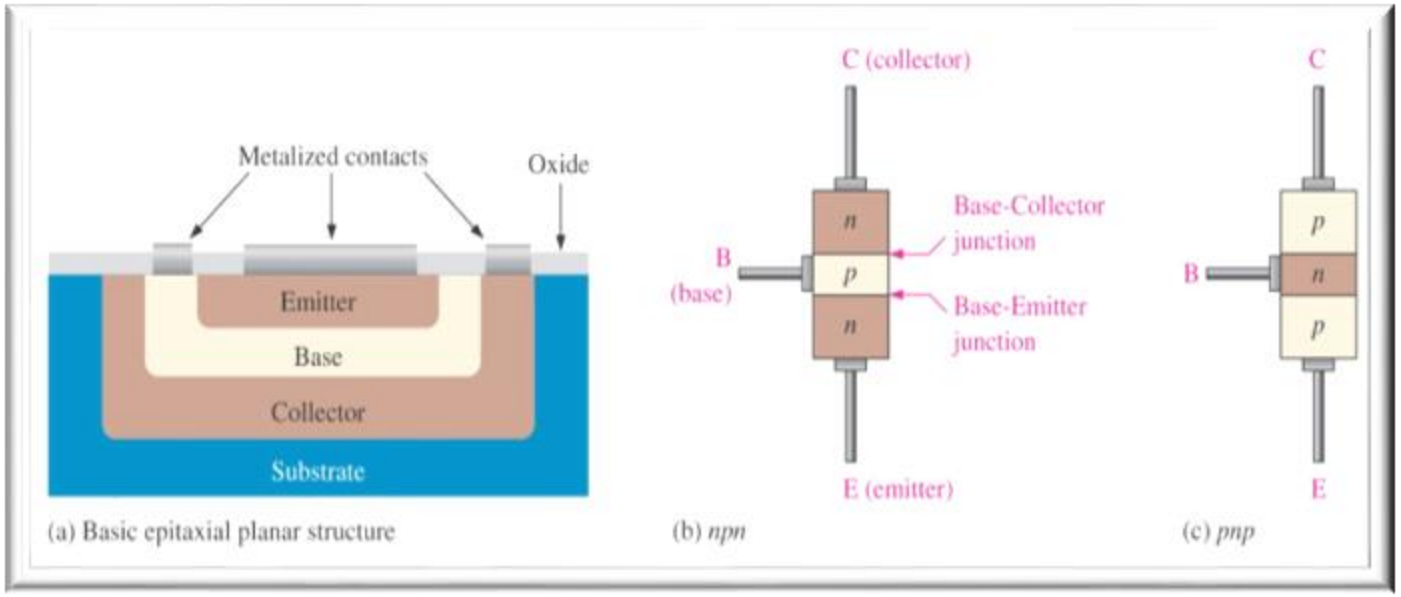
Why BJT is Called Bipolar?

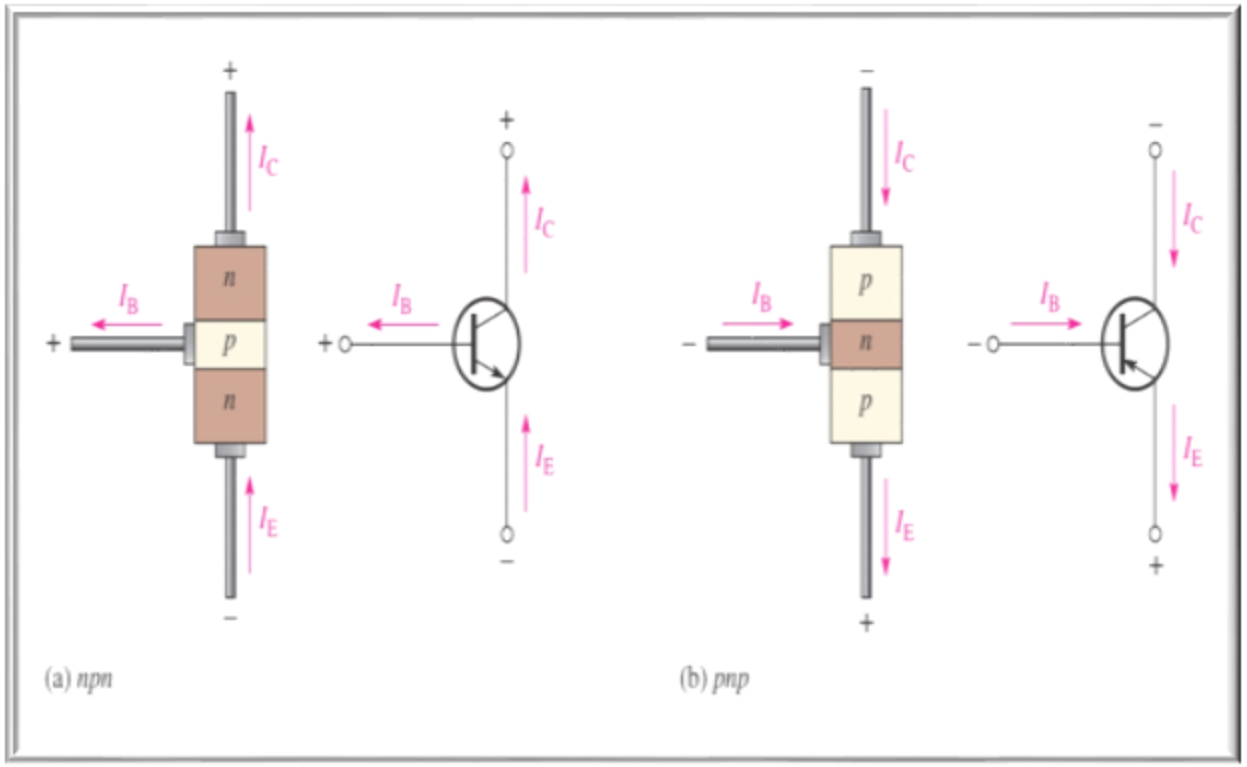
The basic of electronic system nowadays is semiconductor device.

A BJT is a solid-state device in which the current flow between two terminals (the collector and the emitter) is controlled by the amount of current that flows through a third terminal (the base).

BJT consists of three terminal:

- Collector : C
- Base : B
- Emitter : E





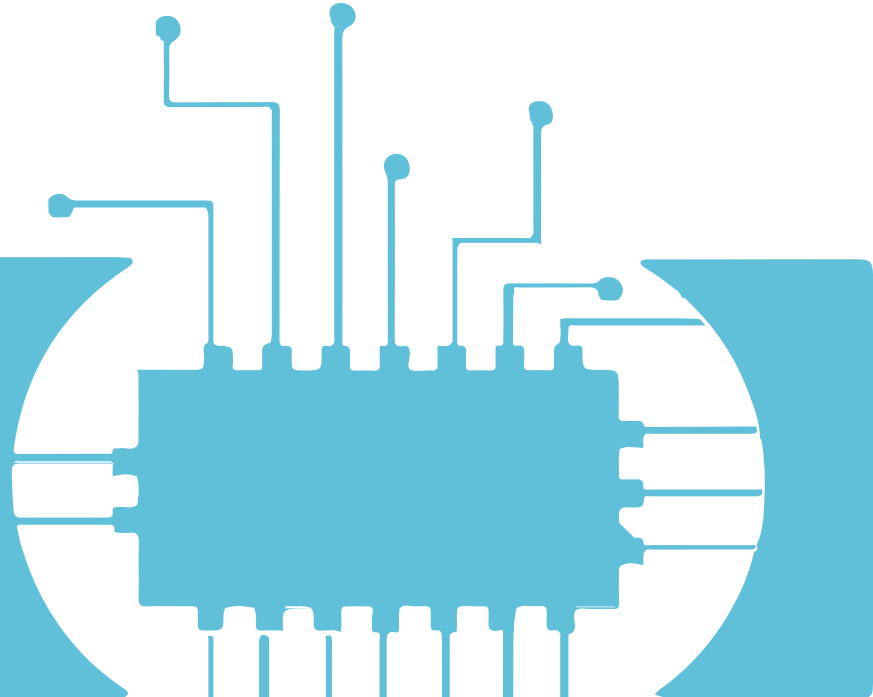
A BJT is one of the most common types of transistors, and can be either NPN or PNP.

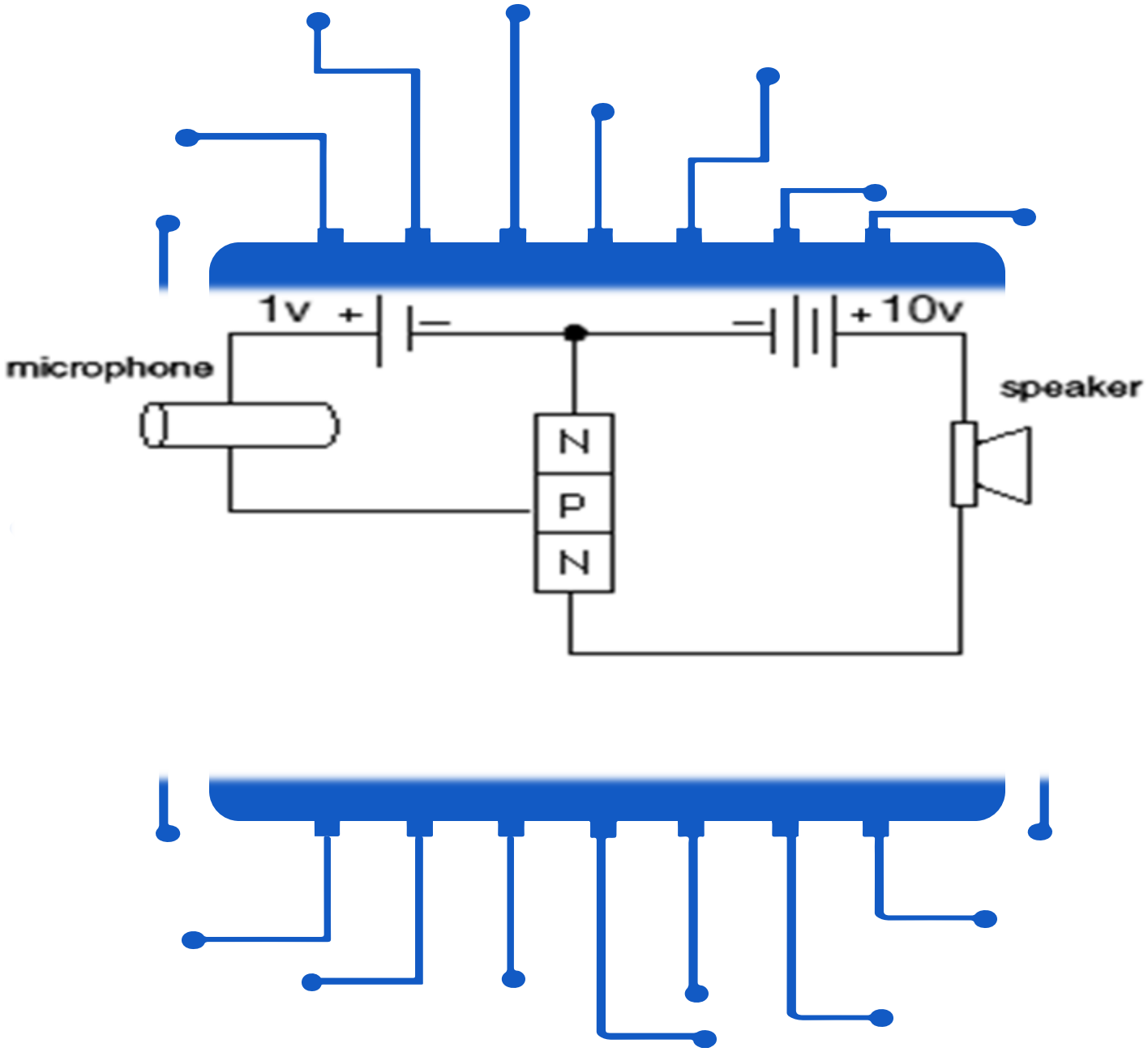
By joining these three layers, a BJT can amplify an electrical signal or switch the current on or off.

Bipolar transistors are so named because the controlled current must go through *two* types of semiconductor material: P and N. The current consists of both electron and hole flow, in different parts of the transistor.

Bipolar transistors consist of either a P-N-P or an N-P-N semiconductor “sandwich” structure.

Architecture of BJTs





Simulation of transistor as an amplifier

Transistor Examples in Daily Life

Microphones

Hearing Aids

Air Conditioner



Compact Disc Players

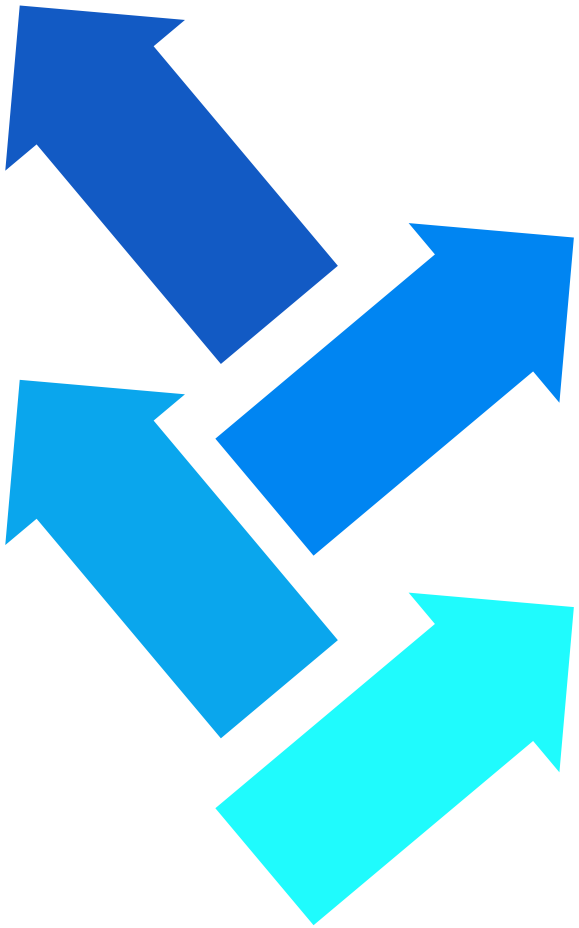
Computer Processors

Astronomy

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